《coker像素级模仿美赛o奖论文指南第6篇》如何画出o奖论文的图表

> 💡 《coker像素级模仿美赛O奖论文指南》
>
> 我的目标是帮助更多人通过模仿对标的方式快速熟悉某个事物或者入门一项技能。我的服务会随着我的成长而不断迭代，我希望能和大家共同成长。
>
> 目前我的身份是数学专业的大学生。对于与我类似的，比我慢几步的人来说，美赛的指导是我能提供的服务之一。
>
> 因此便有了这篇文档。这篇文档最初发布在微信群。
>
> 如果你在其他地方看到了这篇文档，欢迎关注我。
>
> 目前我在所有的平台都叫:@coker无限成长。

美赛又被称为美术大赛

那么到底要画多少图呢？应该画那些图呢？评委喜欢怎么样的图呢？

我们可以从o奖论文中进行模仿学习

在此之前，除了用软件直接生成图外，一定要学会利用ppt来绘图。

ppt比ps多一个p，所以p图的效率和效果显然会更好。

先说结论:

- 数量上，一共要有20张左右的图表，其中应该有背景部分的插图，our work图，符号图，灵敏性分析图。其余为模型建立部分的插图
- 模型建立和结果部分的图，尽量要做到风格统一，同时，尽量以长条状为主(可以是两张图/三张图并排放)(换句话说不要只放一张像正方形一样的图在中间)。尽量不要超过1/2页面，最好是1/4到1/3
- 图片的内容主要是: 结果的展示，以及**结果(假设)的检验(或者与实际情况的对比)，**后者容易被忽略，但是其实更重要。从比例上来说两类的数量差不多。

- 可以对软件生成的图片进行进一步加工，例如在背景用不同颜色作区分，加入一些符号等。
- 尝试使用雷达图或弦图
- 表格可以加颜色

具体统计结果和论文中的图片请继续往下看

24年o奖论文的图表数如下:

A题	B题	C题
20图+3表=23	13图+10表=23	16图+3表=19
17图+1表=18	15图+2表=17	15图+6表=21
18图+4表=22	15图+1表=16	12图+4表=16
21图+3表=24	19图+3表=22	18图+6表=24
24图+6表=30		14图+3表=17
D题:	E题	15图+1表=16
16图+5表=21	11图+4表=15	15图+6表=21
19图+4表=23	13图+9表=22	16图+3表=19
10图+5表=15	18图+5表=23	14图+5表=19
12图+3表=15	8图+4表=12	18图+13表=31
	13图+2表=15	21图+5表=26
	10图+4表=14	

F题

15图+6表=21, 14图+7表=21,15图+3表=18, 14图+3表=17,17图+12表=29

根据统计结果来看，图和表在20个左右比较合适，考虑到摘要，目录，以及memo等，实际上这意味着大概正文每一页一张图/表(这是数据上而不是排版上)

下面继续分析之前选出来的三个论文的配图和表格。

已经分享过的图片和表格不再放出。

注意:**图下表上**

即图名在图片下，表名在表格上

C题

背景，our work各配图1张，层次分析法过程，指标图 两张。

相关系数热力图:

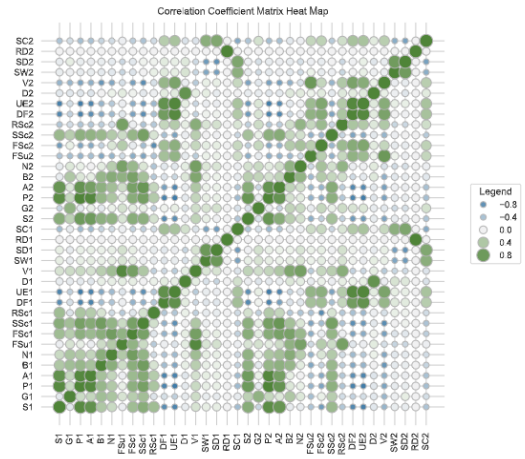


Figure 5: Heat Map of Correlation

评价结果用折线图

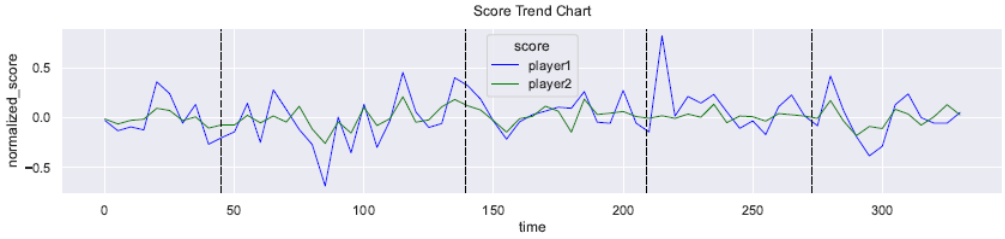


Figure 7: Score Trend Chart

自相关检验

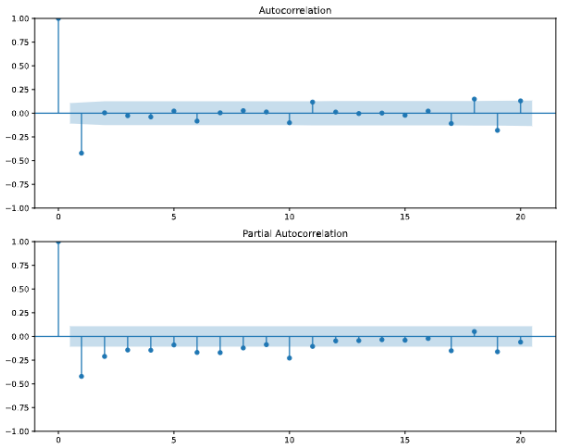


Figure 8: Autocorrelation Graph and Partial Autocorrelation Graph

测试集检验结果

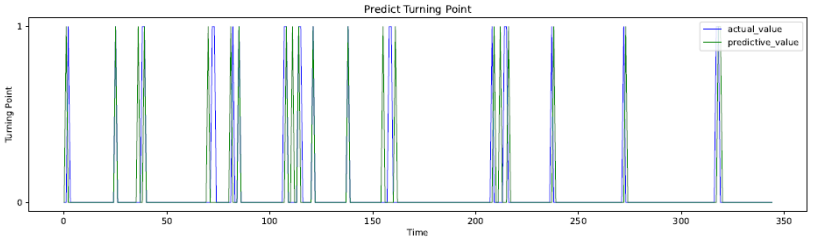


Figure 9: Predictive and Actual Turning plot in Gentlemen's Singles Final

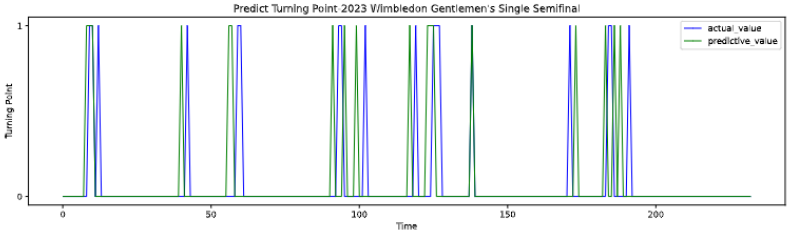


Figure 10: Predictive and Actual Turning Points in Gentlemen's Singles Semifinal

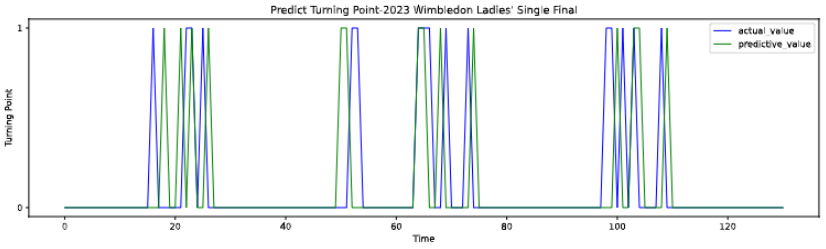


Figure 11: Predictive and Actual Turning Points in Ladies' Singles Final

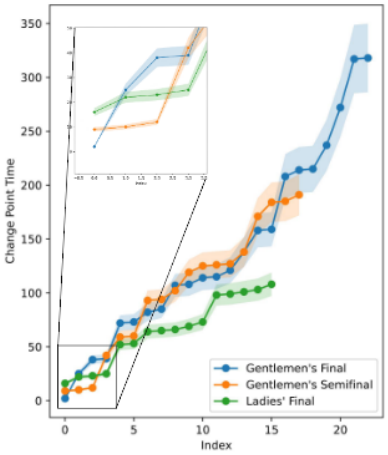


Figure 12: Contrast of Three Matches

指标相关性"

Table 3: Correlation of 18 Indicators

r_S	r_G	r_P	r_A	r_B	r_N	r_{FSu}	r_{FSc}	r_{SSc}
0.9244	0.9250	0.8330	0.9238	0.9239	0.9239	0.9246	0.9237	0.9239
r_{RSc}	r_{DF}	r_{UE}	r_D	r_V	r_{SW}	r_{SD}	r_{RD}	r_{SC}
0.9237	−0.9238	−0.9008	0.6744	0.8993	0.8043	0.8846	0.8866	0.7953

灵敏性分析:

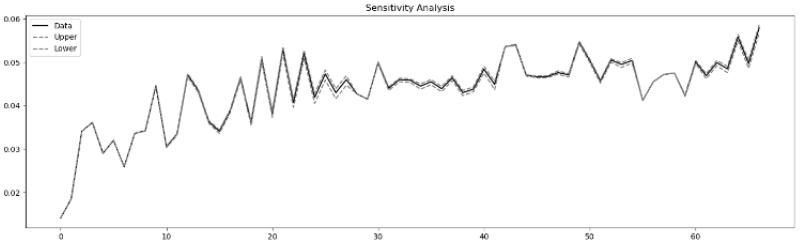


Figure 13: Sensitivity Analysis

> 这里要关注到的核心是: 一致性，从得分趋势图，到测试集检验结果，再到灵敏性分析，这里图片的风格(从形状上来看)是一致的。并且对测试集的检验，延续了3个section，三个分别检验而且最终又放到了一张图来展示，非常连贯。

E题:

同样的，背景部分，our work各有一张图。notion和data 各有一张表。

检验数据的分布:

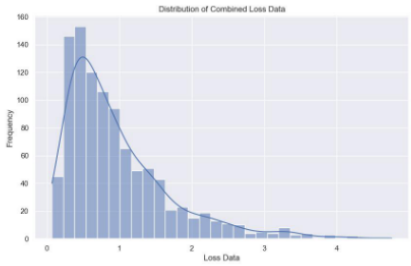


Figure 5: Frequency Histogram and Fitting Effect of Expanded Samples

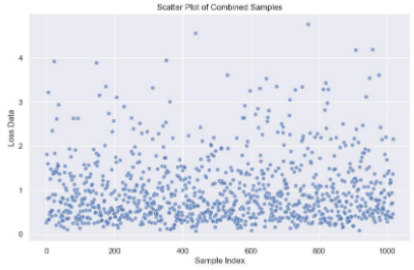


Figure 6: Scatter Plot of the Expanded Samples

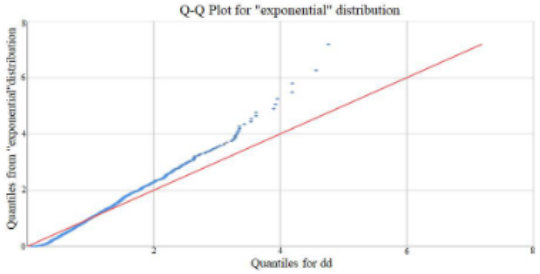


Figure 7: Q-Q plots on the Standard Exponential Distribution

Table 3: Fitting Results of the Frequency of Extreme Weather Events

	K-S static (Mumbai)	p-value (Mumbai)	K-S static (Cairns)	p-value (Cairns)
Normal Distribution	0.186	0.03	0.277	<0.01
Uniform Distribution	2.041	<0.01	2.449	<0.01
Poisson Distribution	0.662	0.77	0.874	0.43
Exponential Distribution	2.393	<0.01	2.399	<0.01

检验结果与应用:

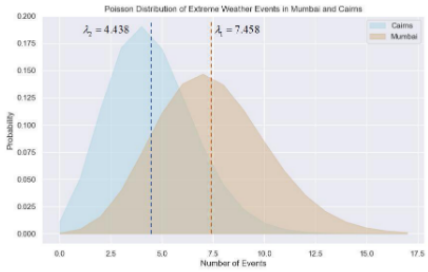


Figure 8: Probability Density Function Graph for Two Sites

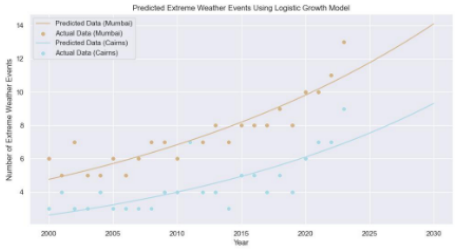


Figure 9: Prediction of Annual Frequency of Extreme Weather Events

分级与实证分析

Table 4: Risk Level for Insurance Companies

Range of POR	<2%	2%~5%	5%~12%	>12%
Risk Level	I (Acceptable)	II (Tolerable)	III (Cautious)	IV (Unacceptable)

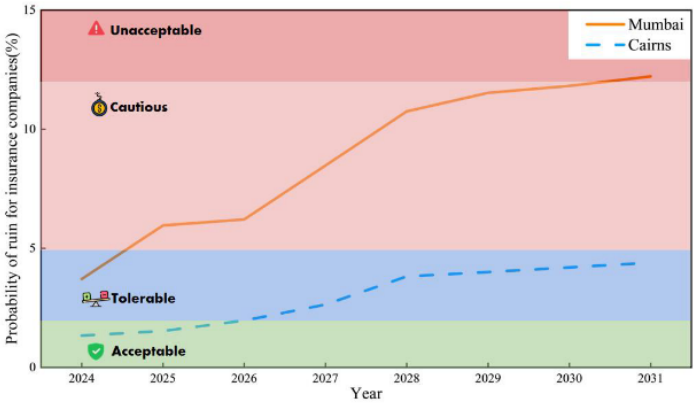


Figure 10: Predictions of POR in Mumbai and Cairns

图片进行了后期加工。

层次分析法的图片展示（可见上一篇)

模型检验:

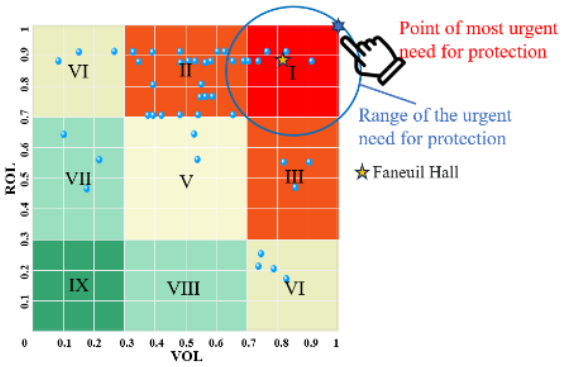


Figure 16: GE Matrix Diagram

此图仍然被后期加工过

实证研究:

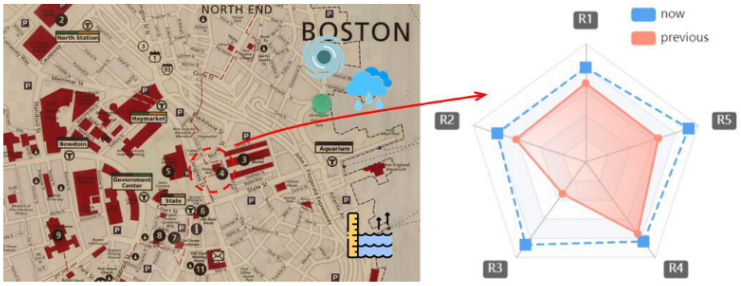
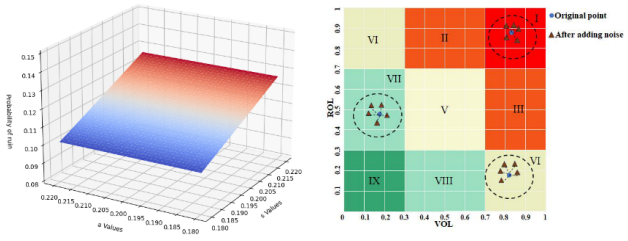


Figure 17: Comparison of Risks Faced by Faneuil Hall in Last Five Years

灵敏性分析



(a) Sensitivity Analysis of Climate Risk Assess Model (b) Robust Test of Landmark Preservation Model

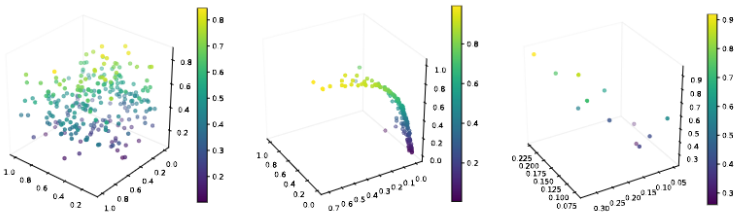
Figure 18: Result of Sensitivity Analysis and Robust Test

F题:

背景部分插入+our work+notation表格

以及层次分析法部分内容(见前篇)

展示结果:



(a) government (b) enterprise (c) organization

Figure 4: Three-dimensional profile data vector visualization

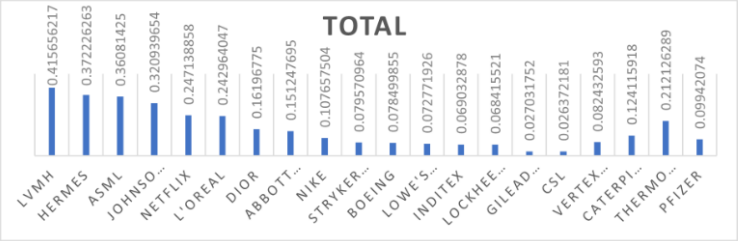


Figure 5: Comparison of total scores for some typical enterprises (just few samples for illustration)

(a) Scores of Power (b) Scores of Resources (c) Scores of Interests (d) Scores of Total

Figure 6: Scores comparison of some enterprises (just few samples for illustration)

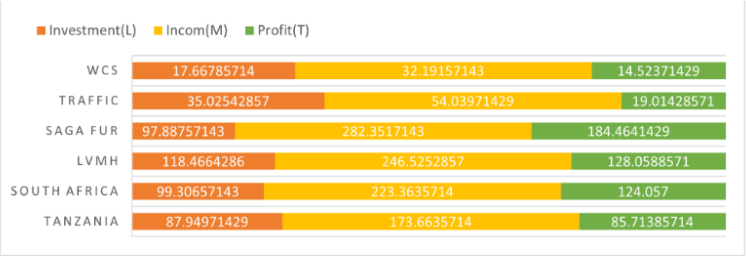


Figure 7: Expected profits results of samples

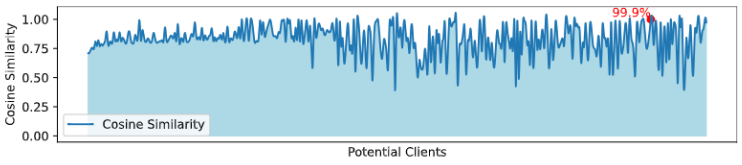
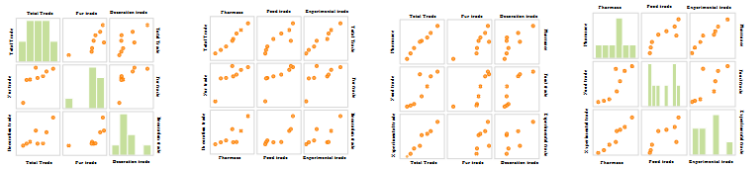


Figure 8: The most suitable entity for augmenting our client

非法贸易相关性散点图



(a) Total & Fur & Decoration (b) Pharmance & Food & Experimental (c) Total & Fur & Decoration (d) Pharmance & Food & Experimental

Figure 9: A scatter plot of the total illegal trade in animals and plants and its associated factors

线性回归结果:

Table 3: Linear Regression Analysis Results

	β	t	P	VIF	R^2
Illegal Fur Trade	-0.147	-0.158	-0.042	0.507	12.88
Illegal Food Trade	0.164	0.075	0.039	0.729	11.725
Illegal Medical Trade	0.812	1.01	0.027	0.396	14.092
Illegal Decoration Trade	-1.811	-0.03	-0.025	0.823	4.65
Illegal Experimental trade	2.651	0.068	0.003	0.927	14.175

残差检验:

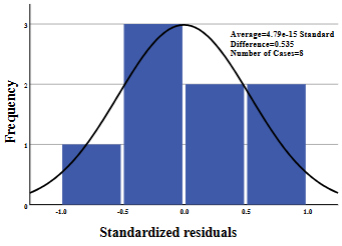


Figure 10: Residual Histogram

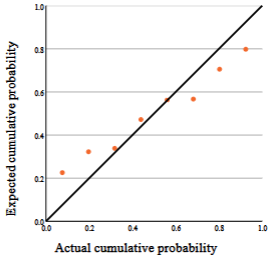


Figure 11: P-P Plot of Residuals

实际值与预测值，影响因素弦图

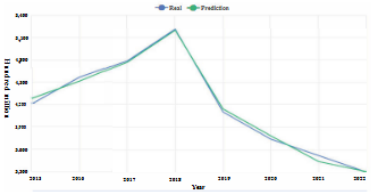


Figure 12: Actual vs. Predicted Values Plot

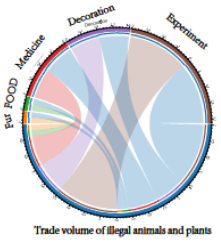
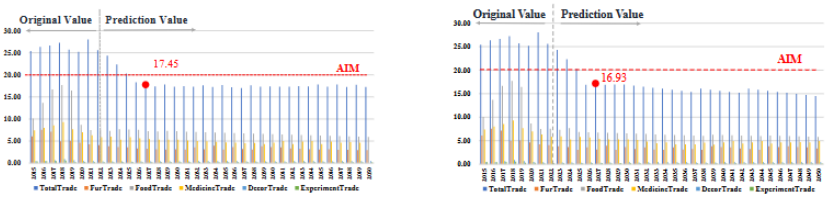


Figure 13: Influencing factors and chord diagrams

敏感性分析



(a) Before adjustment (b) After adjustment

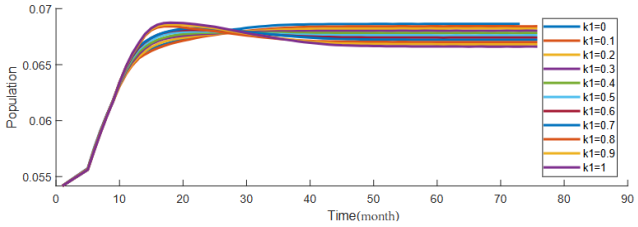


Figure 15: Population dynamics

可以看到其实都是类似的

这里也很明显看到了ppt的痕迹，即一些标记等。

此外也都是长条状，内容主要是模型的结果和对假设的检验，大约一半一半。